BODY—BODY JOINT REINFORCEMENT MATERIAL (STRUCTURAL FOAM)—REPAIR PROCEDURE	Article No. 02-17-9

FORD: 2003 EXPEDITION

LINCOLN: 2003 NAVIGATOR

ISSUE

The following document outlines repair procedures for collision-damaged vehicles that contain Structural Foam.

ACTION

Inspect collision-damaged sheet metal for Structural Foam and replace it following the repair procedure below.

BACKGROUND

Structural Foam

Structural Foam is a new technology material, which is being used in vehicles to reinforce joints, body panels and cavities. This material is not the same as Acoustical Foams, Sound Deadeners, and Hem Adhesives. Structural Foams are significantly different from any other foam product currently on the market. There are two types of Structural Foam applications that are being used in vehicles today.

Carrier Application

The first application type is referred to as a carrier application. This application uses a thin layer of Structural Foam attached to a steel or nylon reinforcement. These parts are then put into the vehicle in locations such as A-pillars, D-pillars, Frame Rails, and Underbody Joints. When these parts go through the OEM E-coat baking process the Structural Foam expands and the result is a joint that is reinforced from the inside where traditional welding would not reach. The 2003 Expedition/Navigator Structural Foam applications are of this type.

Full-Fill Application

The second application type is referred to as a full-fill application. In this application, a two-part epoxy Structural Foam is pumped into selected locations in the body sheet metal or frame that require reinforcing. The result is a fill of cement-like material that adheres to the metal and creates a solid joint. The epoxy cures at room temperature and does not require high temperature ovens.

SERVICE PROCEDURE

NOTE

READ THIS REPAIR PROCEDURE COMPLETELY BEFORE PERFORMING ANY SERVICE ACTIONS.

Structural Foam is being used in the following six locations on the Expedition and Navigator vehicles.

Note: Right and Left as seated in vehicle.

- Upper B-pillar - Right and Left sides
- Upper D-pillar - Right and Left sides
- Lower D-pillar - Right and Left sides

Whether body parts are being replaced or repaired you must do one of the following:

- If sheet metal is being replaced in any of these mentioned locations you will need to fully fill the cavities.
- If (repairing not replacing body sheet metal) the structural foam parts appear to be damaged you must full-fill each area.

This will encase reinforcement parts in place and eliminate rattles. The repair procedures should be done after sheet metal work has been completed and before final painting of the vehicle.

Cartridge Get Ready Procedure (X)

Use Motorcraft (TA-12) Structural Foam to fill the cavities of the vehicle that need to be repaired. Structural Foam can be applied using either a manual or pneumatic application gun. Foam flow can be significantly affected by temperature and will flow easily when it reaches a temperature range of 115°F to 165°F (46°-74 °C). It is highly recommended that the Structural Foam cartridges be heated before use. Put the cartridges in a paint booth during a bake cycle or place the cartridges in a bucket of hot water for approximately 20 minutes.

CAUTION

ONCE HEATED, THE CARTRIDGE IS HOT AND GLOVES MAY BE REQUIRED FOR HANDLING.

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Once the foam is heated, place it in the applicator gun and dispense a small amount until the material is fully mixed inside the mixing nozzle. The material is now ready to be applied.

Upper B-Pillar - Right And Left Sides (Figure 1)

This application will require 3 cartridges of foam

1. Cut a 3/4″ wide 4″ X 3″ piece of foam.
2. Insert foam into the shoulder harness anchor location in B-Pillar. Push foam up so it expands across pillar.
3. Cut two 1 1/2″ wide 4″ X 3″ pieces of foam.
4. Insert one piece into the hole in center of the rear door side rail.
5. Insert the other piece into the hole in the rear of the A-pillar header.
6. Apply aluminum duct tape to all holes above upper D-ring shoulder harness location.
7. Grease the shoulder harness anchor bolt and insert into the shoulder harness anchor location.
8. Follow cartridge get ready procedure (X).
9. Insert cartridge into applicator and dispense a small amount of foam to insure proper flow. Attach mixing tip and run a bead the length of the mixer.
10. Insert mixer into upper most hole of center B-pillar.
11. Dispense all three cartridges.

Upper D-Pillar - Right Or Left Side (Figure 2)

This application will require 14 cartridges of foam

1. Remove the pear shaped plastic plug in the upper D-pillar.
2. Cut and insert a 5.5x4x3 inch (140x100x75mm) soft foam block into the upper D-pillar through the pear shaped hole and slide the block downward past the rubber plug to the taped over round hole.
3. Cut and Insert a 5.5x4x3 inch (140x100x75mm) soft foam block into the upper D-pillar through the pear shaped hole and slide the block upward toward the front of the vehicle.
4. Reinstall the pear shaped plug.
5. Remove the round rubber plug.
6. Apply aluminum duct tape to the tab hole and (8 mm diameter) holes in the rear of the vehicle.
7. Follow the foam get ready procedure (X) to begin application.
8. Dispense the foam into the upper D-pillar through the round hole (rubber plugged). Allow the foam to cure for 60 minutes and remove aluminum tape as necessary.

Lower D-Pillar - Right Or Left Side (Figure 3)

This application will require 14 cartridges of foam

1. Insert a 5.5x4x3 inch (140x100x75mm) soft foam block into the #4 body mount area (oval hole inside the lift gate area). The front edge of the block should be outboard toward the D-pillar. The block will prevent foam expansion into the body mount.
2. Apply aluminum duct tape to cover all holes on the inside of the body and under side of the body at the repair area.
3. Follow the foam get ready procedure (X) to begin application.
4. Dispense the foam into the lower D-pillar at the bottom through the large rectangular hole in the taillight opening.
5. Allow the foam to cure for 60 minutes and remove aluminum tape as necessary.

NOTE

STRUCTURAL FOAM KIT CONTAINS FOUR CARTRIDGES OF STRUCTURAL FOAM AND ONE FOAM BLOCK.

Vehicle Location	Structural Foam Cartridge Requirements	Foam Block Requirements
Upper B-Pillar	3	1
Upper D-Pillar	14	1
Lower D-pillar	14	N/R

PART NUMBER	PART NAME
TA-12	Motorcraft Structural Foam Kit
TA-10	Motorcraft Manual Application Gun
TA-11	Motorcraft Pneumatic Application Gun

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 111000

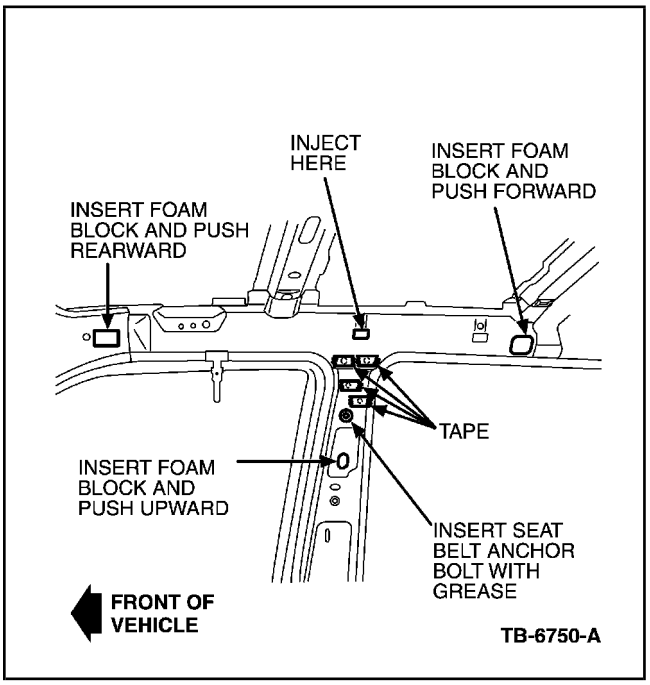


Figure 1 - Article 02-17-9

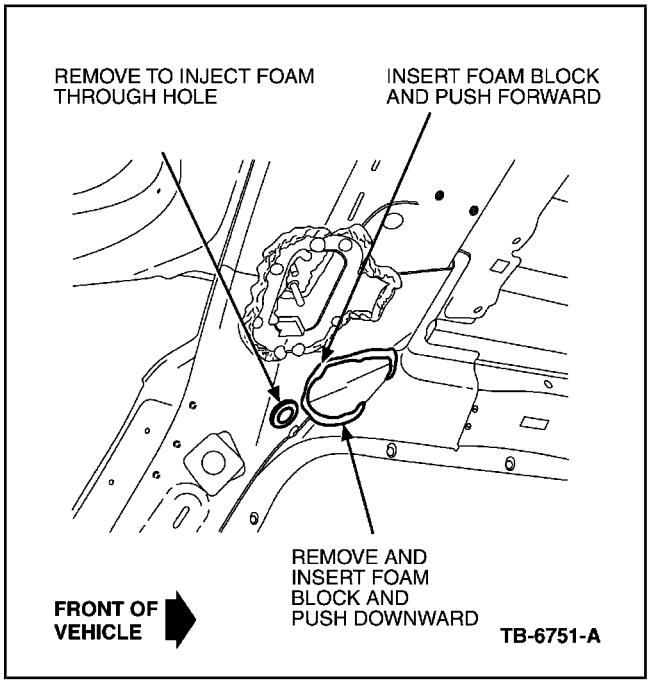


Figure 2 - Article 02-17-9

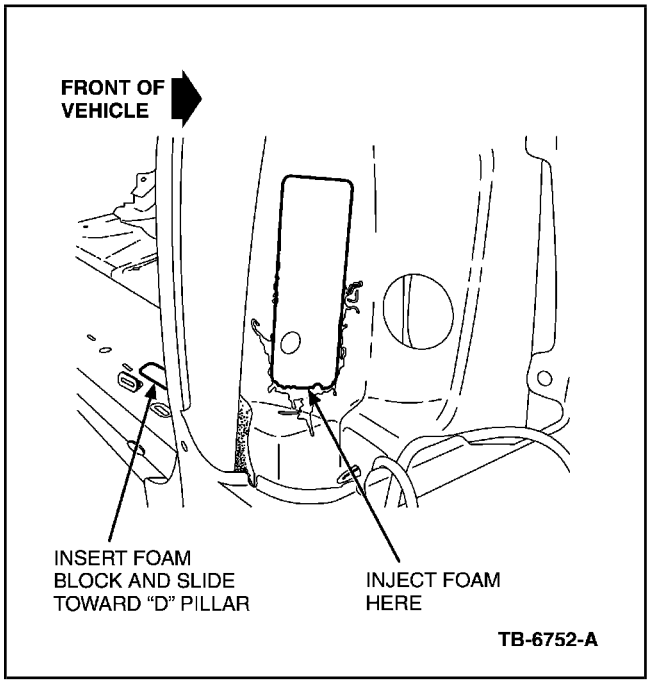


Figure 3 - Article 02-17-9

NOTE: The information in Technical Service Bulletins is intended for use by trained, professional technicians with the knowledge, tools, and equipment to do the job properly and safely. It informs these technicians of conditions that may occur on some vehicles, or provides information that could assist in proper vehicle service. The procedures should not be performed by "do-it-yourselfers". Do not assume that a condition described affects your car or truck. Contact a Ford, Lincoln, or Mercury dealership to determine whether the Bulletin applies to your vehicle.